AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM354
Software Engineering

Midterm Examination (MTA)
Spring Semester 2024/2025

Date: 5 /Apr./2025

Number of Exam Pages: 3 (including this cover sheet)	**Time Allowed:** 2 Hours

Instructions:

- Total Marks: **60**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are **not** allowed.

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Part I – Multiple Choice Questions — 6 Marks

*Answer **all** the following multiple-choice questions. Each question is worth 2 Marks.*

1. Which characteristic of software that affects development is easy to recognize, but difficult to define?
   a. Malleability b. Complexity c. Size d. Usability

2. Which security requirement addresses consistency of data?
   a. access b. privacy c. integrity d. audit

3. Which UML diagram is best used to represent a Business Process:
   a. Activity diagram b. Use Case diagram c. Class diagram d. Object diagram

Part II – Short Answer Questions — 24 Marks

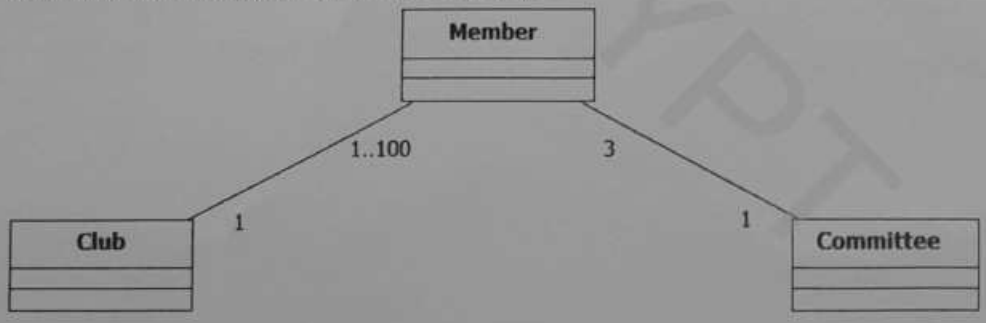
*Answer **all** the following short answer questions. Each question is worth 6 Marks.*

1. a) list 4 of the main characteristics of a good software system.
   b) How might greater flexibility make a software system more affordable over its whole life?

2. a) Briefly discuss maintainability from 2 aspects (adapting to new requirements, finding and fixing errors (bugs)).
   b) How can maintainability be measured in each case?

3. a) What is a fit-criterion for a functional requirement?
   b) Does a fit-criterion indicate how the functional requirement would be tested?

4. Given the following part of a class diagram:



   a) State the meaning of the listed multiplicities
   b) List 2 possible problems with the multiplicity values that are provided

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Part III – Problem Solving Questions — 30 Marks

*Answer **all** the following problem-solving questions. Each question is worth 15 Marks.*

1. Given the below Use Case Diagram for an e-Commerce system
   a. Redraw the diagram using generalization properly, without adding any new actor(s).
   b. Prepare the Use Case description for the "Delete Content" use case.



2. Develop an activity diagram for the task of "Purchasing and Downloading a Console Game". Make sure to cover the following requirements:
   - Use Swimlanes to represent 2 roles, the User and the Console System
   - Use Decision and Merge nodes at least twice each (Hint: You can check for sufficient storage).
   - Make sure to use at least 6 activities

End of questions

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